

Mini-Test
Modern Physics

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Text and solution for Monday's mini-test.

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WEEK VI

- Consider a particle on the $[-1, 1]$ interval with the normalized wavefunction

$$\psi(x) = \frac{\sqrt{15}}{4}(1 - x^2).$$

Compute the expectation value of $\hat{p}^2 = -\hbar^2 \partial^2 / \partial x^2$ in this state (2pt).

Note: The wavefunction in the printed version of the mini test had an *incorrect* normalization. We did not penalize any results obtained with that value, but we provide the solution using the correct value here.

Solution

The expectation value of an operator $\hat{O}$ on a normalized wavefunction is given by

$$\langle \hat{O} \rangle = \int_{-\infty}^{\infty} dx \psi^*(x) \hat{O} \psi(x).$$

In this case

$$\langle \hat{p}^2 \rangle = -\hbar^2 \frac{15}{16} \int_{-1}^1 dx (1 - x^2) \frac{\partial^2}{\partial x^2} (1 - x^2)$$

where the derivative only acts on $\psi(x)$, yielding

$$\langle \hat{p}^2 \rangle = \hbar^2 \frac{15}{8} \int_{-1}^1 dx (1 - x^2) = \hbar^2 \frac{15}{8} \left[x - \frac{x^3}{3} \right]_{-1}^1 = \frac{5}{2} \hbar^2.$$

- Consider a particle on the $[-1, 1]$ interval with the normalized wavefunction

$$\psi(x) = \sqrt{\frac{2}{\pi}} \sin(nx).$$

Compute the expectation value of $\hat{p}^2 = -\hbar^2 \partial^2 / \partial x^2$ in this state (2pt).

Note: There were *two versions* of this mini test, one for $n = 2$ and the other for $n = 3$. We provide the general solution for $n > 0$ here.

Solution

Analogously to the other case, the expectation value is given by

$$\langle \hat{p}^2 \rangle = -\hbar^2 \frac{2}{\pi} \int_0^\pi dx \sin(nx) \frac{\partial^2}{\partial x^2} \sin(nx)$$

where the derivative only acts on $\psi(x)$, yielding

$$\langle \hat{p}^2 \rangle = \hbar^2 \frac{2}{\pi} n^2 \int_0^\pi dx \sin^2(nx).$$

Using the trigonometric identity $\sin^2(nx) = \frac{1}{2} [1 - \cos(2nx)]$ to get

$$\langle \hat{p}^2 \rangle = \hbar^2 \frac{1}{\pi} n^2 \int_0^\pi dx [1 - \cos(2nx)].$$

The first term trivially integrates to π , while the second term is proportional to

$$\left. \frac{1}{2n} \sin(2nx) \right|_0^\pi = 0$$

. So, the final result is

$$\langle \hat{p}^2 \rangle = n^2 \hbar^2.$$